

ENGR 213: Applied Ordinary Differential Equations

Paradis notes — In-Class Lecture, Tutorial & Midterm 1 Preparation Master Compendium

Department of Mechanical, Industrial & Aerospace Engineering · Concordia University

Academic Session: Fall 2026 / Winter 2025 · **Course Code:** ENGR 213
Instructor: Dr. Alexandre Paradis, ing. Ph.D.
Curriculum Mapping: Lectures 1–5 · Tutorials 1 & 3 · Homework Sets 1–3 · Midterm #1 Examination Scope (October 19: Chapter 2 + §17.1 & §17.2)

Executive Blueprint & Pedagogical Architecture

This compendium provides an authoritative, rigorously structured, and comprehensive mathematical synthesis of Dr. Alexandre Paradis’s in-class lectures, weekly tutorials, assigned textbook homework sets, and Midterm #1 preparation. All handwritten derivations and whiteboard notes from the official course documentation have been transformed into publication-grade mathematical typesetting with complete proofs, physical interpretations, verification steps, and high-yield examination strategies.

Curriculum Phase	Topics & Sections Covered	Primary Analytical Competencies
Module 1 (Lecture 1)	§1.1, §1.2: ODE vs. PDE, Order, Degree, Linearity	Classification criteria, Leibniz/Lagrange notations, Linearity Rules
Module 2 (Tutorial 1 & HW 1)	§1.1, §2.1: Solution Verification, Phase Line Stability	Radical domain restrictions, autonomous critical points, attractors vs. repellers
Module 3 (Lecture 2 & HW 2)	§2.2: Separation of Variables, Singular Solutions	Differential separation, lost equilibrium solutions, explicit IVP intervals
Module 4 (Lecture 3 & HW 3)	§2.3: First-Order Linear Equations & Integrating Factors	Standard form normalization, integrating factor $I(x) = e^{\int P(x)dx}$, transient terms
Module 5 (Tutorial 3 & Lecture 4)	§2.4, §2.7: Exact Equations & Physical Models	Test for exactness, potential function $\Psi(x, y)$, radioactive decay, Newton’s cooling
Module 6 (Lecture 4)	§2.5, §2.8: Substitutions & Nonlinear Models	Bernoulli $u = y^{1-n}$, homogeneous $y = ux$, logistic growth, Torricelli’s tank
Module 7 (Lecture 5)	§17.1, §17.2: Complex Variables & Polar Forms	Modulus, principal argument in radians, Euler’s formula, De Moivre’s theorem, n -th roots
Module 8 (Midterm 1 Strategy)	Midterm 1 Examination Synthesis & Pitfalls	10 high-yield exam traps, rapid check algorithms, calculator policy
Module 9 (Get Ready for Midterm 1)	Complete Midterm 1 Scope (§2.1–§2.8, §17.1, §17.2)	Active 7-day revision system, 3-phase diagnostic checklist, and in-depth step-by-step problem guides for every exam topic

Module 1: Fundamental Concepts & Classification (Lecture 1)

Date Delivered: September 8, 2026 · **Textbook Reference:** Zill, Sections 1.1, 1.2 & 2.1

1.1 Course Administration & Policy

- **Approved Calculator Policy:** Concordia University Gina Cody School of Engineering enforces a strict calculator policy. Only standard approved models bearing the official ENCS faculty sticker are permitted in examinations.
- **Midterm #1 Date:** Monday, October 19, 2026.
- **Midterm #1 Scope:** Chapter 2 (Sections 2.1–2.8) + Chapter 17 (Sections 17.1 & 17.2).

1.2 Definition of Differential Equations

A differential equation is any mathematical relation involving one or more dependent variables (unknown functions) and one or more of their derivatives with respect to one or more independent variables.

- **Ordinary Differential Equation (ODE):** Involves derivatives with respect to only a single independent variable (e.g., x or t):

$$\frac{dy}{dx} + y = 0, \quad \left(\frac{dy}{dx}\right)^2 + 2xy = \cos(x), \quad \frac{d^2y}{dt^2} + 4y = 0$$

- **Partial Differential Equation (PDE):** Involves partial derivatives of a multivariable function with respect to two or more independent variables:

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0, \quad \frac{\partial y}{\partial t} = \alpha^2 \frac{\partial^2 y}{\partial x^2}$$

1.3 Standard Derivative Notations

- **Leibniz Notation:** $\frac{dy}{dx}, \frac{d^2y}{dx^2}, \dots, \frac{d^ny}{dx^n}$ (essential for differential operations, separation of variables, and chain rule manipulations).
- **Prime (Lagrange) Notation:** $y', y'', y''', y^{(4)}, \dots, y^{(n)}$ (compact and standard for linear constant-coefficient ODEs).
- **Newton Dot Notation:** $\dot{y} = \frac{dy}{dt}, \ddot{y} = \frac{d^2y}{dt^2}$ (standard in engineering dynamics, kinematics, and vibrations).

1.4 Order and Degree of an ODE

- **Order:** The order of an ODE is defined strictly as the order of the highest derivative present in the equation.
- **Degree:** The power (exponent) to which the highest derivative is raised, after the equation has been rationalized with respect to derivatives.

$\frac{dy}{dx} - y^4 = 0 \implies \text{1}^{\text{st}} \text{ Order, Degree 1}$ (power of y does not alter order)

$\left(\frac{dy}{dx}\right)^3 + y = 0 \implies \text{1}^{\text{st}} \text{ Order, Degree 3}$

$\frac{d^4y}{dx^4} + \frac{d^2y}{dx^2} + 1 = 0 \implies \text{4}^{\text{th}} \text{ Order, Degree 1}$

1.5 Linearity Criteria

An n -th order ODE is **linear** if and only if it can be expressed in the standard canonical form:

$$a_n(x)\frac{d^ny}{dx^n} + a_{n-1}(x)\frac{d^{n-1}y}{dx^{n-1}} + \cdots + a_1(x)\frac{dy}{dx} + a_0(x)y = g(x)$$

The Two Golden Rules of Linearity:

1. **Rule 1 (First Power Only):** The dependent variable y and every single one of its derivatives $y', y'', \dots, y^{(n)}$ appear to the first power only (no terms like $(y')^2$, y^3 , or $\sqrt{y}$).
2. **Rule 2 (Independent Coefficients):** The coefficient functions $a_n(x), \dots, a_0(x)$ and the non-homogeneous driving term $g(x)$ depend strictly and solely on the independent variable x (no cross-products such as yy' or transcendental functions of y like $\sin(y)$ or e^y).

Differential Equation	Order	Linear?	Mathematical Justification
$(y - x)dx + 4xdy = 0$	1	Yes	Dividing by dx : $4xy' + y = x$ satisfies standard linear form.
$y' = y^2 - 3y$	1	No	Nonlinear due to the quadratic power y^2 .
$y'' + 4y = \sin(x)$	2	Yes	Dependent variable y and y'' are linear; $\sin(x)$ depends only on x .
$y'' + 4 \sin(y) = 0$	2	No	Nonlinear due to transcendental function $\sin(y)$ of the dependent variable.
$x^2y''' + xy' - 5y = e^x$	3	Yes	3rd-order linear with variable coefficients $x^2, x, -5$.

Module 2: In-Class Tutorial 1 & Homework 1 Solutions

Date Delivered: September 14, 2026 · Focus: Solution Verification, Radical Domains & Autonomous Phase Portraits

2.1 Problem §1.1 Q13: First-Order Linear Verification

- **Given ODE & Trial Solution:**

$2y' + y = 0, \quad y(x) = e^{-x/2}$

- **Step-by-Step Analytical Derivation:**

1. Compute the first derivative using the exponential chain rule:

$$y'(x) = \frac{d}{dx} \left[e^{-x/2} \right] = -\frac{1}{2}e^{-x/2}$$

2. Substitute $y(x)$ and $y'(x)$ into the left-hand differential operator:

$$\text{LHS} = 2y' + y = 2 \left(-\frac{1}{2}e^{-x/2} \right) + e^{-x/2} = -e^{-x/2} + e^{-x/2} = 0$$

3. Since $\text{LHS} = 0 = \text{RHS}$, the function satisfies the ODE identically for all $x \in \mathbb{R}$.

- **Interval of Definition:**

$$I = (-\infty, +\infty)$$

2.2 Problem §1.1 Q17: Radical Solution & Domain Restriction

- **Given ODE & Trial Solution:**

$(y - x)y' = y - x + 8, \quad y(x) = x + 4\sqrt{x + 2}$

- **Step-by-Step Analytical Derivation:**

1. Differentiate the trial function:

$$y'(x) = \frac{d}{dx} [x + 4(x+2)^{1/2}] = 1 + 4 \left(\frac{1}{2\sqrt{x+2}} \right) = 1 + \frac{2}{\sqrt{x+2}}$$

2. Evaluate the Left-Hand Side (LHS):

$$y - x = (x + 4\sqrt{x+2}) - x = 4\sqrt{x+2}$$
$$\text{LHS} = (y - x)y' = (4\sqrt{x+2}) \left(1 + \frac{2}{\sqrt{x+2}} \right) = 4\sqrt{x+2} + \frac{8\sqrt{x+2}}{\sqrt{x+2}} = 4\sqrt{x+2} + 8$$

3. Evaluate the Right-Hand Side (RHS):

$$\text{RHS} = y - x + 8 = (x + 4\sqrt{x+2}) - x + 8 = 4\sqrt{x+2} + 8$$

4. Notice $\text{LHS} \equiv \text{RHS}$ identically.

- **Interval of Validity Determination:** The square root $\sqrt{x+2}$ requires $x+2 \geq 0 \implies x \geq -2$. However, in the derivative $y'(x) = 1 + \frac{2}{\sqrt{x+2}}$, the term $\sqrt{x+2}$ appears in the denominator, requiring $x+2 \neq 0 \implies x > -2$.

$$\mathbf{I} = (-2, +\infty)$$

2.3 Problem §1.1 Q25: Second-Order Homogeneous Verification

- **Given ODE & Trial Solution:**

$$y'' - 4y' + 4y = 0, \quad y(x) = c_1 e^{2x} + c_2 x e^{2x}$$

- **Step-by-Step Analytical Derivation:**

1. Write the trial function in factored form: $y(x) = (c_1 + c_2 x) e^{2x}$.
2. Compute the first derivative via product rule:

$$y'(x) = c_2 e^{2x} + 2(c_1 + c_2 x) e^{2x} = (2c_1 + c_2 + 2c_2 x) e^{2x}$$

3. Compute the second derivative:

$$y''(x) = 2c_2 e^{2x} + 2(2c_1 + c_2 + 2c_2 x) e^{2x} = (4c_1 + 4c_2 + 4c_2 x) e^{2x}$$

4. Substitute into the differential operator:

$$y'' - 4y' + 4y = e^{2x} [(4c_1 + 4c_2 + 4c_2 x) - 4(2c_1 + c_2 + 2c_2 x) + 4(c_1 + c_2 x)]$$
$$= e^{2x} [(4c_1 - 8c_1 + 4c_1) + (4c_2 - 4c_2) + (4c_2 x - 8c_2 x + 4c_2 x)] = e^{2x} [0] \equiv 0$$

- **Interval of Definition:** $I = (-\infty, +\infty)$.

2.4 Problem §2.1 Q21: Autonomous Direction Fields & Stability

- **Given Autonomous ODE:**

$$\frac{dy}{dx} = y^2 - 3y = y(y - 3)$$

- **Critical Points (Equilibrium Solutions):** Set $\frac{dy}{dx} = 0 \implies y(y - 3) = 0 \implies y_1 = 0$ and $y_2 = 3$.
- **Phase Line Sign Analysis:**
 - **Interval $(-\infty, 0)$:** Test $y = -1 \implies (-1)(-4) = +4 > 0 \implies \frac{dy}{dx} > 0$ (arrows point UP toward $y = 0$).
 - **Interval $(0, 3)$:** Test $y = 1 \implies (1)(-2) = -2 < 0 \implies \frac{dy}{dx} < 0$ (arrows point DOWN toward $y = 0$).
 - **Interval $(3, +\infty)$:** Test $y = 4 \implies (4)(1) = +4 > 0 \implies \frac{dy}{dx} > 0$ (arrows point UP away from $y = 3$).
- **Stability Classification:**
 - $y = 0$ is **Asymptotically Stable (Attractor / Sink)**: Nearby solution trajectories converge to 0 as $x \rightarrow +\infty$.
 - $y = 3$ is **Unstable (Repeller / Source)**: Nearby solution trajectories diverge away from 3 as $x \rightarrow +\infty$.

Module 3: Separable Differential Equations & IVPs (Lectures 2 & 3)

Dates Delivered: September 10 & 22, 2026 · **Textbook Reference:** Zill, Section 2.2

3.1 Mathematical Theory of Separable Equations

A first-order differential equation is **separable** if the right-hand side can be factored into a product of a function of x and a function of y :

$$\frac{dy}{dx} = g(x)f(y)$$

Standard 4-Step Analytical Algorithm:

1. **Check for Singular Solutions:** Set $f(y) = 0$. Any constant root $y = c$ is an equilibrium solution. Check at the end whether it is included in the general family.
2. **Separate Differentials:** Divide by $f(y)$ and multiply by dx :

$$\frac{1}{f(y)}, dy = g(x), dx \quad (f(y) \neq 0)$$

3. **Integrate Both Sides:**

$$\int \frac{1}{f(y)}, dy = \int g(x), dx + C$$

4. **Isolate $y(x)$ and Apply Initial Conditions:** Use initial condition $y(x_0) = y_0$ to isolate C and explicitly define the solution's interval of definition.

3.2 Dr. Paradis In-Class Quiz 1 Master Problem

- **Problem Statement:** Solve the initial value problem:

$$y \frac{dy}{dx} + x^2 = 4, \quad y(0) = 2$$

- **Complete Step-by-Step Derivation:**

1. Isolate the derivative term:

$$y \frac{dy}{dx} = 4 - x^2$$

2. Separate variables:

$$y, dy = (4 - x^2), dx$$

3. Integrate both sides:

$$\int y, dy = \int (4 - x^2), dx \implies \frac{y^2}{2} = 4x - \frac{x^3}{3} + C_1$$

4. Multiply by 2:

$$y^2 = 8x - \frac{2}{3}x^3 + C$$

5. Apply initial condition $y(0) = 2$:

$$(2)^2 = 8(0) - \frac{2}{3}(0)^3 + C \implies 4 = C$$

6. Substitute $C = 4$:

$$y^2 = 8x - \frac{2}{3}x^3 + 4$$

7. Since the initial condition specifies $y(0) = +2 > 0$, we select the positive branch:

$$y(x) = \sqrt{8x - \frac{2}{3}x^3 + 4}$$

- **Interval of Definition:** The solution is valid on the open interval containing $x = 0$ where the radicand remains strictly positive: $8x - \frac{2}{3}x^3 + 4 > 0$.

3.3 Problem 52.2 Q17: Algebraic Factorization Separable ODE

- **Given ODE:**

$$\frac{dy}{dx} = \frac{xy + 3x - y - 3}{xy - 2x + 4y - 8}$$

- **Analytical Derivation:**

1. Factor numerator and denominator by grouping:

$$\text{Numerator} = x(y + 3) - (y + 3) = (x - 1)(y + 3)$$

$$\text{Denominator} = x(y - 2) + 4(y - 2) = (x + 4)(y - 2)$$

2. Rewrite the separated equation:

$$\frac{y - 2}{y + 3}, dy = \frac{x - 1}{x + 4}, dx$$

3. Perform polynomial division on both sides:

$$\frac{y-2}{y+3} = \frac{(y+3)-5}{y+3} = 1 - \frac{5}{y+3}$$
$$\frac{x-1}{x+4} = \frac{(x+4)-5}{x+4} = 1 - \frac{5}{x+4}$$

4. Integrate both sides:

$$\int \left(1 - \frac{5}{y+3}\right) dy = \int \left(1 - \frac{5}{x+4}\right) dx$$
$$y - 5 \ln |y+3| = x - 5 \ln |x+4| + C$$

5. Rearrange terms:

$$y - x + C = 5 \ln \left| \frac{y+3}{x+4} \right|$$

Module 4: First-Order Linear Equations & Integrating Factors (Lecture 3 & HW 3)

Dates Delivered: September 22 & 25, 2026 · **Textbook Reference:** Zill, Section 2.3

4.1 Standard Canonical Form

A first-order linear differential equation must always be written with the leading coefficient equal to +1:

$$\frac{dy}{dx} + P(x)y = Q(x)$$

4.2 The Integrating Factor Theorem

Multiply the entire equation by an unknown strictly positive integrating factor $I(x)$:

$$I(x) \frac{dy}{dx} + I(x)P(x)y = I(x)Q(x)$$

We demand that the left-hand side matches the derivative of the product $[I(x)y]$:

$$\frac{d}{dx}[I(x)y] = I(x) \frac{dy}{dx} + I'(x)y$$

Equating coefficients of y :

$$I'(x) = I(x)P(x) \implies \frac{dI}{I} = P(x), dx \implies \ln |I| = \int P(x), dx \implies \mathbf{I(x) = e^{\int P(x), dx}}$$

4.3 General Solution Formula

Integrating both sides of $\frac{d}{dx}[I(x)y] = I(x)Q(x)$:

$$I(x)y = \int I(x)Q(x), dx + C \implies \mathbf{y(x) = \frac{1}{I(x)} \left[\int I(x)Q(x), dx + C \right]}$$

[!!IMPORTANT] **Dr. Paradis Exam Golden Rule:** The arbitrary constant $+C$ MUST be placed inside the brackets before dividing by $I(x)$. Writing $y = \frac{1}{I(x)} \int IQ dx + C$ is an automatic deduction of full method marks!

4.4 Transient Terms vs. Steady-State Solutions

The general solution can be partitioned into:

$$y(x) = y_c(x) + y_p(x) = \frac{C}{I(x)} + \frac{1}{I(x)} \int I(x)Q(x), dx$$

- **Transient Term:** Any component of $y(x)$ that decays to zero as $x \rightarrow +\infty$ (typically terms with e^{-kx} for $k > 0$):

$$\lim_{x \rightarrow \infty} y_{\text{transient}}(x) = 0$$

- **Steady-State Term:** The persistent long-term behavior of the system as $x \rightarrow +\infty$.

4.5 Problem §2.3 Q21: Singular Points and Interval of Definition

- Given Equation:

$$x \frac{dy}{dx} + 2y = 3, \quad x > 0$$

- Step-by-Step Derivation:

1. Divide by x to achieve standard form:

$$\frac{dy}{dx} + \frac{2}{x}y = \frac{3}{x} \implies P(x) = \frac{2}{x}, \quad Q(x) = \frac{3}{x}$$

2. Compute integrating factor:

$$I(x) = \exp\left(\int \frac{2}{x} dx\right) = e^{2 \ln|x|} = e^{\ln(x^2)} = x^2$$

3. Apply product rule:

$$\frac{d}{dx}[x^2 y] = x^2 \left(\frac{3}{x}\right) = 3x$$

4. Integrate both sides:

$$x^2 y = \int 3x, dx + C = \frac{3}{2}x^2 + C \implies \mathbf{y}(\mathbf{x}) = \frac{\mathbf{3}}{\mathbf{2}} + \frac{\mathbf{C}}{\mathbf{x}^2}$$

5. The term $\frac{C}{x^2}$ is the **transient term** since $\lim_{x \rightarrow \infty} \frac{C}{x^2} = 0$. The **steady-state solution** is $y_{ss} = \frac{3}{2}$.

6. The point $x = 0$ is a **singular point** where $P(x)$ and $Q(x)$ are discontinuous. Thus, the solution exists on $I = (0, +\infty)$.

Module 5: Exact Equations & Physical Modeling (Tutorial 3 & Lecture 4)

Dates Delivered: September 28 & 29, 2026 · **Textbook Reference:** Zill, Sections 2.4, 2.7 & 2.8

5.1 Test for Exactness (§2.4)

A first-order differential expression in differential form:

$$M(x, y), dx + N(x, y), dy = 0$$

is an **exact differential** in a simply connected region R if and only if:

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

5.2 Construction of Potential Function $\Psi(x, y)$

When exactness holds, there exists a potential function $\Psi(x, y)$ such that $\frac{\partial \Psi}{\partial x} = M$ and $\frac{\partial \Psi}{\partial y} = N$.

1. Integrate $M(x, y)$ with respect to x , introducing an arbitrary function of integration $g(y)$:

$$\Psi(x, y) = \int M(x, y), dx + g(y)$$

2. Differentiate $\Psi(x, y)$ with respect to y and set equal to $N(x, y)$:

$$\frac{\partial \Psi}{\partial y} = \frac{\partial}{\partial y} \left[\int M(x, y), dx \right] + g'(y) = N(x, y)$$

3. Isolate $g'(y)$ (verify all x variables cancel out) and integrate to find $g(y)$.

4. The implicit solution is:

$$\Psi(\mathbf{x}, \mathbf{y}) = \mathbf{C}$$

5.3 Special Integrating Factors for Non-Exact Equations

If $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$, compute the test ratios:

- Function of x alone:

$$\frac{M_y - N_x}{N} = f(x) \implies \mu(\mathbf{x}) = \mathbf{e}^{\int \mathbf{f}(\mathbf{x}), d\mathbf{x}}$$

- **Function of y alone:**

$$\frac{N_x - M_y}{M} = g(y) \implies \mu(y) = e^{\int g(y) dy}$$

5.4 Tutorial 3 Worked Problem 1: Radioactive Decay (§2.7)

- **Problem Statement:** An initial radioactive mass $A(0) = 100$ mg decreases by 3 after 6 hours. Find the remaining mass at $t = 24$ hours.
- **Analytical Derivation:**
 1. Decay model: $\frac{dA}{dt} = -kA \implies A(t) = A_0 e^{-kt} = 100e^{-kt}$.
 2. At $t = 6$ h, 3 has decayed, so 97 remains:

$$A(6) = 100e^{-6k} = 97 \implies e^{-6k} = 0.97 \implies -k = \frac{1}{6} \ln(0.97)$$

3. At $t = 24 = 4 \times 6$ hours:

$$A(24) = 100e^{-24k} = 100(e^{-6k})^4 = 100(0.97)^4 \approx \mathbf{88.53 \text{ mg}}$$

5.5 Tutorial 3 Worked Problem 2: Forensic Newton's Law of Cooling (§2.7)

- **Problem Statement:** A body is discovered in a room maintained at constant ambient temperature $T_m = 70^\circ\text{F}$. At time of discovery t_d , the body temperature is 85°F . One hour later ($t_d + 1$), the temperature drops to 80°F . Assuming living body temperature was $T(0) = 98.6^\circ\text{F}$, estimate the time of death.
- **Analytical Derivation:**
 1. Newton's Cooling ODE:

$$\frac{dT}{dt} = -k(T - 70) \implies \int \frac{dT}{T - 70} = -k \int dt \implies T(t) = 70 + Ce^{-kt}$$

2. At time of death $t = 0$:

$$T(0) = 70 + C = 98.6 \implies C = 28.6 \implies T(t) = 70 + 28.6e^{-kt}$$

3. At discovery t_d :

$$85 = 70 + 28.6e^{-kt_d} \implies 28.6e^{-kt_d} = 15 \implies e^{-kt_d} = \frac{15}{28.6}$$

4. One hour later at $t_d + 1$:

$$80 = 70 + 28.6e^{-k(t_d+1)} \implies 28.6e^{-kt_d}e^{-k} = 10$$

5. Substitute $28.6e^{-kt_d} = 15$:

$$15e^{-k} = 10 \implies e^{-k} = \frac{10}{15} = \frac{2}{3} \implies k = -\ln\left(\frac{2}{3}\right) \approx 0.4055 \text{ hr}^{-1}$$

6. Calculate elapsed time t_d :

$$-kt_d = \ln\left(\frac{15}{28.6}\right) \implies t_d = \frac{\ln(15/28.6)}{\ln(2/3)} = \frac{-0.6453}{-0.4055} \approx \mathbf{1.59 \text{ hours (1 hr 35 min)}}$$

- **Conclusion:** Death occurred approximately 1 hour and 35 minutes prior to discovery.

Module 6: Solutions by Substitution & Nonlinear Models (Lectures 4 & 5)

Dates Delivered: September 29 & October 1, 2026 · **Textbook Reference:** Zill, Sections 2.5 & 2.8

6.1 Bernoulli Differential Equations

A first-order equation of the form:

$$\frac{dy}{dx} + P(x)y = Q(x)y^n \quad (n \neq 0, 1)$$

is transformed into a linear ODE via the substitution:

$$u = y^{1-n} \implies \frac{du}{dx} = (1-n)y^{-n} \frac{dy}{dx}$$

Dividing the original ODE by y^n :

$$y^{-n} \frac{dy}{dx} + P(x)y^{1-n} = Q(x) \implies \frac{du}{dx} + (1-n)P(x)u = (1-n)Q(x)$$

6.2 Homogeneous Differential Equations of Degree Zero

An ODE $M(x, y)dx + N(x, y)dy = 0$ where M and N are homogeneous functions of the exact same degree can be written as:

$$\frac{dy}{dx} = f\left(\frac{y}{x}\right)$$

Substitute:

$$y = ux \implies \frac{dy}{dx} = u + x \frac{du}{dx}$$

Substituting into the ODE yields an immediately separable equation:

$$u + x \frac{du}{dx} = f(u) \implies \frac{du}{f(u) - u} = \frac{dx}{x}$$

6.3 Linear Arguments: $y' = f(Ax + By + C)$

When the derivative depends strictly on a linear combination $Ax + By + C$ ($B \neq 0$), substitute:

$$u = Ax + By + C \implies \frac{du}{dx} = A + B \frac{dy}{dx} \implies \frac{dy}{dx} = \frac{1}{B} \left(\frac{du}{dx} - A \right)$$

Separated form:

$$\frac{du}{A + Bf(u)} = dx$$

6.4 Nonlinear Physical Models (§2.8)

- **Logistic Population Growth:**

$$\frac{dP}{dt} = P(a - bP) = aP \left(1 - \frac{P}{K} \right), \quad K = \frac{a}{b} \text{ (Carrying Capacity)}$$

Analytical solution via partial fractions or Bernoulli substitution ($n = 2$):

$$P(t) = \frac{K}{1 + \left(\frac{K - P_0}{P_0} \right) e^{-at}}$$

- **Torricelli's Law (Draining Tank):**

$$\frac{dV}{dt} = -A_h v_{\text{exit}} = -A_h \sqrt{2gh}$$

For a cylindrical container with constant top surface area A_w :

$$A_w \frac{dh}{dt} = -A_h \sqrt{2gh} \implies \frac{dh}{\sqrt{h}} = -\frac{A_h \sqrt{2g}}{A_w} dt$$

Module 7: Complex Numbers & Polar Coordinates (Lectures 4 & 5)

Dates Delivered: September 29 & October 1, 2026 · **Textbook Reference:** Zill, Chapter 17 (Sections 17.1 & 17.2)

7.1 Cartesian Representation (§17.1)

A complex number $z \in \mathbb{C}$ is defined as an ordered pair of real numbers (x, y) :

$$z = x + iy \quad (i = \sqrt{-1}; i^2 = -1)$$

- **Real Part:** $\text{Re}(z) = x$
- **Imaginary Part:** $\text{Im}(z) = y$ (note: $\text{Im}(z)$ is the real coefficient y , NOT iy)
- **Complex Conjugate:** $\bar{z} = x - iy$
- **Fundamental Modulus Identity:**

$$z\bar{z} = (x + iy)(x - iy) = x^2 - (iy)^2 = x^2 + y^2 = |z|^2$$

- **Division by Conjugate Rationalization:**

$$\frac{z_1}{z_2} = \frac{z_1 \bar{z}_2}{z_2 \bar{z}_2} = \frac{(x_1 + iy_1)(x_2 - iy_2)}{x_2^2 + y_2^2} = \frac{(x_1 x_2 + y_1 y_2) + i(x_2 y_1 - x_1 y_2)}{x_2^2 + y_2^2}$$

7.2 Polar & Exponential Representation (\$17.2)

- **Modulus (Length):**

$$r = |z| = \sqrt{x^2 + y^2}$$

- **Argument (Phase Angle):**

$$\theta = \arg(z) \implies \tan \theta = \frac{y}{x}$$

[!WARNING] **Dr. Paradis Golden Rule on Angles:** All complex arguments in ENGR 213 MUST be computed in **radians**, never degrees ($\pi \text{ rad} = 180^\circ$).

- **Principal Argument** $\text{Arg}(z) = \Theta \in (-\pi, \pi]$:

$$\Theta = \begin{cases} \arctan(y/x), & x > 0 \\ \arctan(y/x) + \pi, & x < 0, y \geq 0 \\ \arctan(y/x) - \pi, & x < 0, y < 0 \end{cases}$$

- **Euler's Formula & Polar Form:**

$$e^{i\theta} = \cos \theta + i \sin \theta \implies \mathbf{z} = \mathbf{r}(\cos \theta + i \sin \theta) = \mathbf{r}e^{i\theta}$$

7.3 Arithmetic Operations in Polar Coordinates

Let $z_1 = r_1 e^{i\theta_1}$ and $z_2 = r_2 e^{i\theta_2}$:

1. **Multiplication:**

$$z_1 z_2 = r_1 r_2 e^{i(\theta_1 + \theta_2)} = r_1 r_2 [\cos(\theta_1 + \theta_2) + i \sin(\theta_1 + \theta_2)]$$

(Moduli multiply; arguments add)

2. **Division:**

$$\frac{z_1}{z_2} = \frac{r_1}{r_2} e^{i(\theta_1 - \theta_2)} = \frac{r_1}{r_2} [\cos(\theta_1 - \theta_2) + i \sin(\theta_1 - \theta_2)]$$

(Moduli divide; arguments subtract)

3. **De Moivre's Theorem:**

$$z^n = (r e^{i\theta})^n = r^n e^{in\theta} = \mathbf{r}^n [\cos(\mathbf{n}\theta) + i \sin(\mathbf{n}\theta)]$$

4. **n -th Roots of Complex Numbers:** The n distinct roots $w_k = z^{1/n}$ of $z = r e^{i\theta}$ are given by:

$$\mathbf{w}_k = \sqrt[n]{r} \exp\left(i \frac{\theta + 2k\pi}{n}\right) = \sqrt[n]{r} \left[\cos\left(\frac{\theta + 2k\pi}{n}\right) + i \sin\left(\frac{\theta + 2k\pi}{n}\right) \right]$$

for $k = 0, 1, 2, \dots, n - 1$.

Module 8: Midterm #1 Strategic Preparation & Pitfall Elimination (October 19)

8.1 Ten High-Yield Examination Traps & Checklists

1. **Trap 1: Coefficient Normalization in Linear ODEs:** Never calculate $P(x)$ until the coefficient of $\frac{dy}{dx}$ has been divided out to equal +1.
2. **Trap 2: Bracketing the Integrating Factor:** The arbitrary constant $+C$ belongs strictly inside the brackets: $y = \frac{1}{I(x)} \left[\int I Q dx + C \right]$.
3. **Trap 3: Absolute Values in Logarithms:** $\int \frac{1}{x} dx = \ln |x| \implies e^{\ln |x|} = |x|$. Specify whether $x > 0$ or $x < 0$ to drop the absolute value cleanly.
4. **Trap 4: Lost Singular Solutions:** When separating $\frac{dy}{f(y)} = g(x)dx$, immediately check if $f(y) = 0$ yields constant solutions that are not covered by any value of C .
5. **Trap 5: Exactness Cross-Derivative Signs:** Exactness requires $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$. Do not confuse the partial derivative order.
6. **Trap 6: Correct Bernoulli Exponent:** The substitution is $u = y^{1-n}$, which transforms the equation into $\frac{du}{dx} + (1-n)P(x)u = (1-n)Q(x)$.
7. **Trap 7: Radians vs. Degrees in Complex Arithmetic:** Dr. Paradis penalizes degrees on complex polar exam problems. Ensure calculator mode is in **radians**.
8. **Trap 8: Complex Roots Spacing:** When finding n -th roots, always add $\frac{2k\pi}{n}$ for $k = 0, 1, \dots, n - 1$ to capture all n distinct roots evenly distributed on the circle.
9. **Trap 9: Open Intervals of Definition:** Initial value problem solutions must be stated on open intervals ($I = (-2, \infty)$ rather than $[-2, \infty)$) where the derivative is defined.
10. **Trap 10: Sign Conventions in Decay & Cooling:** Rate constant $k > 0$ means decay is written as $\frac{dA}{dt} = -kA$. Cooling is $\frac{dT}{dt} = -k(T - T_m)$.

Module 9: Get Ready for Midterm 1 — In-Depth Topic Mastery, Step-by-Step Problem Walkthroughs & Active Revision System

9.1 Active Revision Architecture & 7-Day Exam Protocol

To achieve mastery in ENGR 213 Midterm 1, passive review must be replaced with structured active retrieval and diagnostic problem-solving. The examination syllabus strictly comprises **Chapter 2 (Sections 2.1–2.8)** and **Chapter 17 (Sections 17.1 & 17.2)**.

1. The 7-Day Precision Revision Timeline

- Day 7 (Qualitative & Separable Foundations — §2.1, §2.2):** Review autonomous equations $\frac{dy}{dx} = f(y)$, 1D phase portraits, attractors, repellers, and semi-stable equilibria. Practice separable ODEs, always checking for lost constant solutions when dividing by $g(y) = 0$.
- Day 6 (Linear First-Order Equations — §2.3):** Normalize to $y' + P(x)y = Q(x)$ before computing integrating factor $I(x) = \exp(\int P(x)dx)$. Practice handling absolute values in logarithms ($e^{\ln|x|} = |x|$), separating transient from steady-state components, and finding intervals of definition bounded by singular points.
- Day 5 (Exact Equations & Special Integrating Factors — §2.4):** Test exactness via $\partial M/\partial y = \partial N/\partial x$. Build the potential function $\Psi(x, y) = C$ methodically. Practice identifying special integrating factors $\mu(x) = \exp(\int \frac{M_y - N_x}{N} dx)$ and $\mu(y) = \exp(\int \frac{N_x - M_y}{M} dy)$.
- Day 4 (Substitutions & Reductions — §2.5):** Test homogeneity of degree k and substitute $y = ux$ or $x = vy$. Master Bernoulli ODE reduction $u = y^{1-n}$ and linear combination shifts $u = Ax + By + C$.
- Day 3 (Physical Modeling & Initial Value Applications — §2.7, §2.8):** Formulate variable-volume mixture tanks $\frac{dA}{dt} = R_{in} - R_{out}$, Newton's Law of Cooling $\frac{dT}{dt} = -k(T - T_m)$, Torricelli's tank draining, and logistic growth $dN/dt = rN(1 - N/K)$.
- Day 2 (Complex Numbers & Polar Forms — §17.1, §17.2):** Master Cartesian conjugate division, exact polar conversion ($r = |z|, \Theta = \text{Arg}(z) \in (-\pi, \pi]$ strictly in radians), Euler's formula $e^{i\theta}$, De Moivre's theorem $z^n = r^n e^{in\theta}$, calculating all n distinct roots w_k , and solving second-order auxiliary equations with complex roots.
- Day 1 (Full-Length Timed Simulation):** Complete an authentic past midterm (e.g. Winter 2025 Term Test 1) under closed-book, timed (75 minutes) conditions using only an approved ENCS calculator. Grade strictly and audit mistakes in the Error Log.

2. The 3-Phase Problem Solving Diagnostic Checklist

Before writing your first algebraic line on the midterm exam, run through this mental checklist:

[Phase 1: Classification & Normalization]

- What is the order and degree of the differential equation?
- Is it linear, separable, exact, homogeneous, or Bernoulli?
- Is the leading derivative coefficient strictly normalized to +1?
- Are differentials properly grouped as $M(x,y)dx + N(x,y)dy = 0$?

[Phase 2: Systematic Technique Execution]

- If Separable: Gather all y terms on LHS with dy and all x terms on RHS with dx . Note any division by zero as equilibrium solutions.
- If Linear: Compute $I(x) = \exp(\int P dx)$. Verify product rule derivative on LHS: $d/dx[I(x)y] = I(x)Q(x)$.
- If Exact: Check $\partial M/\partial y = \partial N/\partial x$. If unequal, test $(M_y - N_x)/N$ for $\mu(x)$ or $(N_x - M_y)/M$ for $\mu(y)$.
- If Bernoulli: Divide by y^n , substitute $u = y^{1-n}$, and rewrite as linear ODE in u .
- If Complex: Keep angles strictly in RADIANS. Use $\text{Arg}(z) \in (-\pi, \pi]$.

[Phase 3: Initial Conditions, Validation & Interval of Definition]

- Apply initial conditions carefully (watch negative signs in $t = -1$ or $y = -1$).
- If a square root branch $\pm\sqrt{\dots}$ is taken, explicitly choose the sign matching $y(x_0)$.
- State the open interval of definition I containing the initial point x_0 .
- Quick Differentiation Check: Plug your derived $y(x)$ back into the original ODE.

3. Master Formula & High-Yield Memory Matrix

ODE / Topic Type	Canonical Standard Form	Key Substitution / Integrating Factor	Solution Formula / Operational Core
Separable ODE	$\frac{dy}{dx} = g(x)h(y)$	Separate: $\frac{1}{h(y)}dy = g(x)dx$	$\int \frac{1}{h(y)}dy = \int g(x)dx + C$; check $h(y) = 0$
Linear First-Order	$y' + P(x)y = Q(x)$	$I(x) = e^{\int P(x)dx}$	$y(x) = \frac{1}{I(x)} [\int I(x)Q(x)dx + C]$
Exact Equation	$M(x, y)dx + N(x, y)dy = 0$	Exact if $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$	$\Psi(x, y) = \int Mdx + g(y) = C$, with $g'(y) = N - \frac{\partial}{\partial y} \int Mdx$
Special Factor $\mu(x)$	Non-exact $Mdx + Ndy = 0$	$\frac{M_y - N_x}{N} = f(x)$	$\mu(x) = e^{\int f(x)dx}$
Special Factor $\mu(y)$	Non-exact $Mdx + Ndy = 0$	$\frac{N_x - M_y}{M} = g(y)$	$\mu(y) = e^{\int g(y)dy}$
Homogeneous ODE	$y' = f(y/x)$	$y = ux \implies y' = u + xu'$	$\frac{du}{f(u)-u} = \frac{dx}{x}$
Bernoulli Equation	$y' + P(x)y = Q(x)y^n$	$u = y^{1-n} \implies u' = (1-n)y^{-n}y'$	$\frac{du}{dx} + (1-n)P(x)u = (1-n)Q(x)$ (Linear in u)
Linear Argument	$y' = f(Ax + By + C)$	$u = Ax + By + C \implies u' = A + By'$	$\frac{du}{A+Bf(u)} = dx$
Mixture Tank	$\frac{dA}{dt} = R_{in} - R_{out}$	$R_{in} = c_{in}r_{in}; R_{out} = \frac{A(t)}{V_0+(r_{in}-r_{out})t}r_{out}$	Linear 1st-order ODE solved via integrating factor
Newton's Cooling	$\frac{dT}{dt} = -k(T - T_m)$	Separate or Linear	$T(t) = T_m + (T_0 - T_m)e^{-kt}$
Complex Polar Form	$z = x + iy$	$r = \sqrt{x^2 + y^2}; \theta = \text{Arg}(z) \in (-\pi, \pi]$	$z = r(\cos \theta + i \sin \theta) = re^{i\theta}$
De Moivre Powers	$z^n = (re^{i\theta})^n$	Exponentiate modulus, multiply argument	$z^n = r^n[\cos(n\theta) + i \sin(n\theta)] = r^n e^{in\theta}$

ODE / Topic Type	Canonical Standard Form	Key Substitution / Integrating Factor	Solution Formula / Operational Core
All n -th Complex Roots	$w_k = z^{1/n}$	Space roots by $\frac{2k\pi}{n}$ for $k = 0, \dots, n - 1$	$w_k = \sqrt[n]{r} \left[\cos\left(\frac{\theta+2k\pi}{n}\right) + i \sin\left(\frac{\theta+2k\pi}{n}\right) \right]$
Auxiliary ODE Roots	$ay'' + by' + cy = 0$	Characteristic roots $\lambda = \alpha \pm i\beta$	$y(x) = e^{\alpha x} [C_1 \cos(\beta x) + C_2 \sin(\beta x)]$

4. Self-Assessment Error Audit Log Template

Keep a dedicated error tracking sheet during midterm revision:

[Error Category]	[Specific Mathematical Slip]	[Remediation Strategy]
1. Normalization Slip:	Forgot to divide by leading x^2 .	Always circle y' to verify coefficient is $+1$.
2. Sign Error in IVP:	Evaluated $-5(-1)$ as -5 instead of $+5$.	Use explicit brackets: $-5(-1) = +5$.
3. Radical Branch Choice:	Kept $\pm$ instead of picking unique sign.	Check sign of initial condition $y(x_0)$.
4. Degree Mode Error:	Wrote $\text{Arg}(z) = 120^\circ$ on exam sheet.	Switch calculator to RAD; write $2\pi/3$.
5. Lost Constant Solution:	Divided by $y^2 - 1$ and lost $y = \pm 1$.	Write $y = \pm 1$ explicitly before dividing.

9.2 In-Depth Step-by-Step Topic Mastery & Exemplar Problem Walkthroughs

Topic 1: Autonomous First-Order ODEs, Critical Points & Phase Lines (§2.1)

- Core Theory:** An ODE is autonomous if $\frac{dy}{dx} = f(y)$ has no explicit dependence on the independent variable x . Critical (equilibrium) points satisfy $f(c) = 0$. On a 1D vertical phase line:
 - If $f(y) > 0$ for $y > c$ and $f(y) < 0$ for $y < c$, solutions diverge away: **Repeller (Unstable)**.
 - If $f(y) < 0$ for $y > c$ and $f(y) > 0$ for $y < c$, solutions converge toward c : **Attractor (Asymptotically Stable)**.
 - If $f(y)$ has the same sign on both sides of c , solutions pass through: **Semi-Stable**.
- Exemplar Examination Problem:** Find all critical points, sketch the 1D phase line, classify stability, and determine $\lim_{x \rightarrow \infty} y(x)$ for $\frac{dy}{dx} = y^2(4 - y)(y + 2)$.
- Analytical Step-by-Step Derivation:**

1. Find Equilibrium Points:

$$y^2(4 - y)(y + 2) = 0 \implies y = -2, \quad y = 0, \quad y = 4$$

2. Analyze Derivative Signs on Intervals:

- Interval $(-\infty, -2)$: Test $y = -3 \implies (-3)^2(4 - (-3))(-3 + 2) = (9)(7)(-1) < 0$ ($\frac{dy}{dx} < 0$, downward motion $\downarrow$).
- Interval $(-2, 0)$: Test $y = -1 \implies (-1)^2(4 - (-1))(-1 + 2) = (1)(5)(1) > 0$ ($\frac{dy}{dx} > 0$, upward motion $\uparrow$).
- Interval $(0, 4)$: Test $y = 1 \implies (1)^2(4 - 1)(1 + 2) = (1)(3)(3) > 0$ ($\frac{dy}{dx} > 0$, upward motion $\uparrow$).
- Interval $(4, \infty)$: Test $y = 5 \implies (5)^2(4 - 5)(5 + 2) = (25)(-1)(7) < 0$ ($\frac{dy}{dx} < 0$, downward motion $\downarrow$).

3. Classify Each Critical Point:

- At $y = -2$: Derivatives point away ($\downarrow$ below, $\uparrow$ above) $\implies$ **Repeller / Unstable**.
- At $y = 0$: Derivatives point upward on both sides ($\uparrow$ below, $\uparrow$ above) $\implies$ **Semi-Stable**.
- At $y = 4$: Derivatives point inward ($\uparrow$ below, $\downarrow$ above) $\implies$ **Attractor / Asymptotically Stable**.

4. Long-Term Asymptotic Behavior ($\lim_{x \rightarrow \infty} y(x)$):

- If $y(0) = 5$: $y_0 > 4 \implies \lim_{x \rightarrow \infty} y(x) = 4$.
- If $y(0) = 2$: $0 < y_0 < 4 \implies \lim_{x \rightarrow \infty} y(x) = 4$.
- If $y(0) = -1$: $-2 < y_0 < 0 \implies \lim_{x \rightarrow \infty} y(x) = 0$.
- If $y(0) = -3$: $y_0 < -2 \implies \lim_{x \rightarrow \infty} y(x) = -\infty$.

Topic 2: Separable Differential Equations & Initial Value Problems (§2.2)

- Core Theory:** A separable equation can be expressed as $g(y)dy = f(x)dx$. To solve an IVP: separate, integrate both sides, solve for the constant C before manipulating the expression, and select the unique square root or algebraic branch determined by $y(x_0)$.
- Exemplar Examination Problem:** Solve the initial value problem $\frac{dy}{dx} = \frac{2x+1}{2y-2}$, $y(0) = -1$, and state its explicit domain of validity.
- Analytical Step-by-Step Derivation:**

1. Separate Variables:

$$(2y - 2), dy = (2x + 1), dx$$

2. Integrate Both Sides:

$$\int (2y - 2), dy = \int (2x + 1), dx \implies y^2 - 2y = x^2 + x + C$$

3. **Complete the Square on y :**

$$(y - 1)^2 - 1 = x^2 + x + C \implies (y - 1)^2 = x^2 + x + C_1 \quad (\text{where } C_1 = C + 1)$$

4. **Apply Initial Condition $y(0) = -1$:**

$$(-1 - 1)^2 = 0^2 + 0 + C_1 \implies (-2)^2 = C_1 \implies C_1 = 4$$

$$(y - 1)^2 = x^2 + x + 4$$

5. **Select the Correct Branch:**

$$y - 1 = \pm \sqrt{x^2 + x + 4} \implies y(x) = 1 \pm \sqrt{x^2 + x + 4}$$

Since $y(0) = -1$: $1 - \sqrt{0 + 0 + 4} = 1 - 2 = -1$. We must select the **negative branch**:

$$y(x) = 1 - \sqrt{x^2 + x + 4}$$

6. **Determine Interval of Validity:** The radicand must satisfy $x^2 + x + 4 > 0$. The discriminant is $\Delta = 1^2 - 4(1)(4) = -15 < 0$. Because the parabola opens upward and has no real roots, $x^2 + x + 4 > 0$ for all $x \in \mathbb{R}$.

$$\text{Interval of Definition: } I = (-\infty, +\infty)$$

Topic 3: First-Order Linear Equations & Integrating Factors (§2.3)

- **Core Theory:** Normalize to $y' + P(x)y = Q(x)$. The integrating factor is $I(x) = \exp(\int P(x)dx)$. Multiplying through collapses the LHS into $\frac{d}{dx}[I(x)y] = I(x)Q(x)$. Integrate and divide by $I(x)$.
- **Exemplar Examination Problem:** Solve $(x^2 - 1)\frac{dy}{dx} + 2xy = x(x^2 - 1)$, $y(0) = 3$, and state the interval of validity.
- **Analytical Step-by-Step Derivation:**

1. **Normalize to Canonical Standard Form:** Divide by $x^2 - 1$:

$$y' + \frac{2x}{x^2 - 1}y = x$$

Here $P(x) = \frac{2x}{x^2 - 1}$ and $Q(x) = x$.

2. **Compute Integrating Factor $I(x)$:**

$$\int P(x), dx = \int \frac{2x}{x^2 - 1}, dx = \ln |x^2 - 1|$$

$$I(x) = e^{\ln |x^2 - 1|} = |x^2 - 1|$$

On an interval containing $x = 0$, $x^2 - 1 < 0$, so $|x^2 - 1| = -(x^2 - 1)$. Taking $I(x) = x^2 - 1$ satisfies the product rule directly:

3. **Apply the Product Rule Identity:**

$$\frac{d}{dx} [(x^2 - 1)y] = (x^2 - 1)x = x^3 - x$$

4. **Integrate Both Sides:**

$$(x^2 - 1)y = \int (x^3 - x), dx = \frac{x^4}{4} - \frac{x^2}{2} + C$$

5. **Apply Initial Condition $y(0) = 3$:**

$$(0 - 1)(3) = 0 - 0 + C \implies -3 = C$$

6. **Solve Explicitly for $y(x)$:**

$$(x^2 - 1)y = \frac{x^4}{4} - \frac{x^2}{2} - 3 = \frac{x^4 - 2x^2 - 12}{4}$$

$$y(x) = \frac{x^4 - 2x^2 - 12}{4(x^2 - 1)}$$

7. **Interval of Validity Determination:** The coefficient functions in standard form have discontinuities at $x = \pm 1$. Since the initial point is $x_0 = 0$, the solution interval is strictly the largest open interval containing $x = 0$ that excludes the singular points:

$$\text{Interval of Definition: } I = (-1, 1)$$

Topic 4: Exact Equations & Special Integrating Factors (§2.4)

- **Core Theory:** For $M(x, y)dx + N(x, y)dy = 0$, if $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$, then $\Psi(x, y) = C$ exists. If not exact, test:

$$\frac{M_y - N_x}{N} = f(x) \implies \mu(x) = e^{\int f(x)dx} \quad \text{or} \quad \frac{N_x - M_y}{M} = g(y) \implies \mu(y) = e^{\int g(y)dy}$$

- **Exemplar Examination Problem:** Find the general solution to $(3x^2y + 2xy + y^3), dx + (x^2 + y^2), dy = 0$.
- **Analytical Step-by-Step Derivation:**

1. Test for Exactness:

$$M(x, y) = 3x^2y + 2xy + y^3 \implies \frac{\partial M}{\partial y} = 3x^2 + 2x + 3y^2$$

$$N(x, y) = x^2 + y^2 \implies \frac{\partial N}{\partial x} = 2x$$

Since $M_y \neq N_x$, the equation is not exact.

2. Test for an Integrating Factor:

$$\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} = (3x^2 + 2x + 3y^2) - 2x = 3x^2 + 3y^2 = 3(x^2 + y^2)$$

Divide by $N(x, y)$:

$$\frac{M_y - N_x}{N} = \frac{3(x^2 + y^2)}{x^2 + y^2} = 3 \quad (\text{Pure function of } x)$$

3. Compute Integrating Factor $\mu(x)$:

$$\mu(x) = e^{\int 3, dx} = e^{3x}$$

4. Multiply the ODE by $\mu(x) = e^{3x}$:

$$(3x^2ye^{3x} + 2xye^{3x} + y^3e^{3x}), dx + (x^2e^{3x} + y^2e^{3x}), dy = 0$$

5. Construct the Potential Function $\Psi(x, y)$: Integrate the new $N(x, y)$ with respect to y (algebraically simpler):

$$\Psi(x, y) = \int (x^2e^{3x} + y^2e^{3x}), dy = x^2ye^{3x} + \frac{y^3}{3}e^{3x} + h(x)$$

6. Differentiate with Respect to x and Equate to M :

$$\frac{\partial \Psi}{\partial x} = 2xye^{3x} + 3x^2ye^{3x} + y^3e^{3x} + h'(x)$$

Comparing with the new $M(x, y) = 3x^2ye^{3x} + 2xye^{3x} + y^3e^{3x}$:

$$h'(x) = 0 \implies h(x) = 0$$

7. State the Implicit General Solution:

$$e^{3x} \left(x^2y + \frac{y^3}{3} \right) = C$$

Topic 5: Solutions by Substitution: Bernoulli Equations (§2.5)

- **Core Theory:** A Bernoulli ODE has the form $y' + P(x)y = Q(x)y^n$. Divide by y^n : $y^{-n}y' + P(x)y^{1-n} = Q(x)$. Substitute $u = y^{1-n} \implies u' = (1 - n)y^{-n}y'$, transforming the ODE into a linear equation:

$$\frac{du}{dx} + (1 - n)P(x)u = (1 - n)Q(x)$$

- **Exemplar Examination Problem:** Solve the initial value problem $x \frac{dy}{dx} + y = x^2y^2, y(1) = 1$, and state the explicit interval of definition.
- **Analytical Step-by-Step Derivation:**

1. Normalize to Standard Bernoulli Form: Divide by x :

$$\frac{dy}{dx} + \frac{1}{x}y = xy^2 \quad (n = 2)$$

2. **Apply Bernoulli Substitution:**

$$u = y^{1-2} = y^{-1} \implies \frac{du}{dx} = -y^{-2} \frac{dy}{dx}$$

3. **Transform the Equation:** Multiply the normalized ODE by $-y^{-2}$:

$$-y^{-2} \frac{dy}{dx} - \frac{1}{x} y^{-1} = -x \implies \frac{du}{dx} - \frac{1}{x} u = -x$$

4. **Solve the Linear Equation in $u(x)$:** The integrating factor is:

$$I(x) = e^{\int -\frac{1}{x} dx} = e^{-\ln|x|} = \frac{1}{x} \quad (\text{for } x > 0)$$

Multiply by $I(x)$:

$$\frac{d}{dx} \left[\frac{u}{x} \right] = \frac{1}{x} (-x) = -1$$

Integrate both sides:

$$\frac{u}{x} = -x + C \implies u(x) = -x^2 + Cx$$

5. **Back-Substitute $u = 1/y$:**

$$\frac{1}{y} = Cx - x^2 \implies y(x) = \frac{1}{Cx - x^2}$$

6. **Apply Initial Condition $y(1) = 1$:**

$$1 = \frac{1}{C(1) - 1^2} \implies C - 1 = 1 \implies C = 2$$

$$y(x) = \frac{1}{2x - x^2} = \frac{1}{x(2 - x)}$$

7. **Interval of Validity Determination:** The solution blows up (vertical asymptotes) at $x = 0$ and $x = 2$. Since $x_0 = 1 \in (0, 2)$, the solution is defined on:

$$\text{Interval of Definition: } I = (0, 2)$$

Topic 6: Physical Modeling: Variable-Volume Mixture Tanks (§2.7, §2.8)

- **Core Theory:** For a salt solution in a tank, the mass of salt $A(t)$ satisfies:

$$\frac{dA}{dt} = R_{\text{in}} - R_{\text{out}} = c_{\text{in}} r_{\text{in}} - c_{\text{out}} r_{\text{out}} = c_{\text{in}} r_{\text{in}} - \left(\frac{A(t)}{V(t)} \right) r_{\text{out}}$$

where $V(t) = V_0 + (r_{\text{in}} - r_{\text{out}})t$. When $r_{\text{in}} \neq r_{\text{out}}$, volume varies with time, producing a variable-coefficient linear ODE.

- **Exemplar Examination Problem:** A tank initially contains 100 L of pure water ($A(0) = 0$). Brine containing 2 kg/L of salt flows in at 3 L/min. The thoroughly stirred mixture is pumped out at 2 L/min. Find the amount of salt in the tank when it contains 150 L of liquid.

- **Analytical Step-by-Step Derivation:**

1. **Determine Time-Dependent Volume $V(t)$:**

$$V(t) = V_0 + (r_{\text{in}} - r_{\text{out}})t = 100 + (3 - 2)t = 100 + t$$

The tank reaches 150 L when:

$$100 + t = 150 \implies t = 50 \text{ minutes}$$

2. **Formulate the Differential Equation:**

$$R_{\text{in}} = c_{\text{in}} r_{\text{in}} = (2 \text{ kg/L})(3 \text{ L/min}) = 6 \text{ kg/min}$$

$$R_{\text{out}} = \left(\frac{A(t)}{V(t)} \right) r_{\text{out}} = \frac{A(t)}{100 + t} \cdot 2 = \frac{2}{100 + t} A(t)$$

$$\frac{dA}{dt} = 6 - \frac{2}{100 + t} A \implies \frac{dA}{dt} + \frac{2}{100 + t} A = 6$$

3. Find the Integrating Factor:

$$I(t) = e^{\int \frac{2}{100+t} dt} = e^{2 \ln(100+t)} = (100+t)^2$$

4. Solve via Integration:

$$\begin{aligned} \frac{d}{dt} [(100+t)^2 A] &= 6(100+t)^2 \\ (100+t)^2 A &= \int 6(100+t)^2, dt = 2(100+t)^3 + C \\ A(t) &= 2(100+t) + \frac{C}{(100+t)^2} \end{aligned}$$

5. Apply Initial Condition $A(0) = 0$ (Pure Water):

$$\begin{aligned} 0 &= 2(100) + \frac{C}{100^2} \implies 0 = 200 + \frac{C}{10,000} \implies C = -2,000,000 = -2 \times 10^6 \\ A(t) &= 2(100+t) - \frac{2 \times 10^6}{(100+t)^2} \end{aligned}$$

6. Evaluate Salt Content at $t = 50$ min ($V = 150$ L):

$$\begin{aligned} A(50) &= 2(150) - \frac{2 \times 10^6}{(150)^2} = 300 - \frac{2,000,000}{22,500} = 300 - \frac{800}{9} \approx 300 - 88.89 \\ A(50) &= \frac{1900}{9} \text{ kg} \approx 211.11 \text{ kg} \end{aligned}$$

Topic 7: Complex Numbers: Arithmetic, Conjugates & Polar Forms (§17.1)

- **Core Theory:** A complex number $z = x + iy$ has modulus $r = |z| = \sqrt{x^2 + y^2}$ and principal argument $\Theta = \text{Arg}(z) \in (-\pi, \pi]$ (strictly computed in radians). Division is carried out by multiplying numerator and denominator by the complex conjugate $\bar{z}_2 = x_2 - iy_2$.
- **Exemplar Examination Problem:** Given $z_1 = -2 + 2\sqrt{3}i$ and $z_2 = 1 - i$:
 - (a) Compute $\frac{z_1}{z_2}$ in Cartesian form $a + bi$.
 - (b) Convert z_1 and z_2 to exact polar exponential form $re^{i\theta}$.
 - (c) Use polar forms to evaluate the quotient and verify agreement with part (a).
- **Analytical Step-by-Step Derivation:**

1. Part (a): Cartesian Division by Conjugate Multiplication:

$$\frac{z_1}{z_2} = \frac{-2 + 2\sqrt{3}i}{1 - i} = \frac{(-2 + 2\sqrt{3}i)(1 + i)}{(1 - i)(1 + i)} = \frac{-2 - 2i + 2\sqrt{3}i + 2\sqrt{3}i^2}{1^2 - i^2}$$

Since $i^2 = -1$:

$$\frac{z_1}{z_2} = \frac{-2 - 2\sqrt{3} + i(2\sqrt{3} - 2)}{1 - (-1)} = \frac{-(2 + 2\sqrt{3}) + i(2\sqrt{3} - 2)}{2} = -(1 + \sqrt{3}) + i(\sqrt{3} - 1)$$

2. Part (b): Polar Conversions:

- For $z_1 = -2 + 2\sqrt{3}i$:

$$r_1 = |z_1| = \sqrt{(-2)^2 + (2\sqrt{3})^2} = \sqrt{4 + 12} = \sqrt{16} = 4$$

Since $x = -2 < 0$ and $y = 2\sqrt{3} > 0$ (Quadrant II):

$$\Theta_1 = \arctan\left(\frac{2\sqrt{3}}{-2}\right) + \pi = \arctan(-\sqrt{3}) + \pi = -\frac{\pi}{3} + \pi = \frac{2\pi}{3} \text{ rad}$$

$$z_1 = 4e^{i2\pi/3}$$

- For $z_2 = 1 - i$:

$$r_2 = |z_2| = \sqrt{1^2 + (-1)^2} = \sqrt{2}$$

Since $x = 1 > 0$ and $y = -1 < 0$ (Quadrant IV):

$$\Theta_2 = \arctan\left(\frac{-1}{1}\right) = -\frac{\pi}{4} \text{ rad}$$

$$z_2 = \sqrt{2}e^{-i\pi/4}$$

3. Part (c): Polar Division & Verification:

$$\frac{z_1}{z_2} = \frac{4e^{i2\pi/3}}{\sqrt{2}e^{-i\pi/4}} = \frac{4}{\sqrt{2}}e^{i(\frac{2\pi}{3}-(-\frac{\pi}{4}))} = 2\sqrt{2}e^{i(\frac{8\pi+3\pi}{12})} = 2\sqrt{2}e^{i\frac{11\pi}{12}}$$

Converting back to Cartesian using exact trigonometric values:

$$\begin{aligned}\cos\left(\frac{11\pi}{12}\right) &= -\frac{\sqrt{6}+\sqrt{2}}{4}, \quad \sin\left(\frac{11\pi}{12}\right) = \frac{\sqrt{6}-\sqrt{2}}{4} \\ \text{Re} &= 2\sqrt{2}\left(-\frac{\sqrt{6}+\sqrt{2}}{4}\right) = -\frac{2\sqrt{12}+4}{4} = -\frac{4\sqrt{3}+4}{4} = -(1+\sqrt{3}) \quad \checkmark \\ \text{Im} &= 2\sqrt{2}\left(\frac{\sqrt{6}-\sqrt{2}}{4}\right) = \frac{2\sqrt{12}-4}{4} = \frac{4\sqrt{3}-4}{4} = \sqrt{3}-1 \quad \checkmark\end{aligned}$$

Topic 8: De Moivre's Powers, Roots & Second-Order Complex Auxiliary IVPs (§17.2)

• Core Theory:

- **De Moivre's Theorem:** $z^n = r^n[\cos(n\theta) + i\sin(n\theta)] = r^n e^{in\theta}$.
- **The n -th Roots:** The n distinct roots of $z = re^{i\theta}$ are $w_k = \sqrt[n]{r} \exp\left(i\frac{\theta+2k\pi}{n}\right)$ for $k = 0, 1, \dots, n-1$.
- **Auxiliary ODE Roots:** If $a\lambda^2 + b\lambda + c = 0$ yields complex conjugate roots $\lambda = \alpha \pm i\beta$, the general solution to $ay'' + by' + cy = 0$ is:

$$y(x) = e^{\alpha x}(C_1 \cos(\beta x) + C_2 \sin(\beta x))$$

- **Exemplar Problem 1 (All Complex Roots):** Find all 3 cube roots of $z = -8i$ and state each root in Cartesian form $a + bi$.

• Analytical Step-by-Step Derivation:

1. Express $z = -8i$ in polar form:

$$\begin{aligned}r &= |-8i| = 8, \quad \theta = \text{Arg}(-8i) = -\frac{\pi}{2} \text{ rad} \\ z &= 8e^{-i\pi/2}\end{aligned}$$

2. Apply the root formula for $n = 3$:

$$w_k = \sqrt[3]{8} \exp\left(i\frac{-\pi/2 + 2k\pi}{3}\right) = 2 \exp\left(i\left(-\frac{\pi}{6} + \frac{2k\pi}{3}\right)\right) \quad \text{for } k = 0, 1, 2$$

3. Compute each root explicitly:

- For $k = 0$:

$$\theta_0 = -\frac{\pi}{6} \implies w_0 = 2 \left[\cos\left(-\frac{\pi}{6}\right) + i \sin\left(-\frac{\pi}{6}\right) \right] = 2 \left(\frac{\sqrt{3}}{2} - \frac{1}{2}i \right) = \sqrt{3} - i$$

- For $k = 1$:

$$\theta_1 = -\frac{\pi}{6} + \frac{2\pi}{3} = \frac{3\pi}{6} = \frac{\pi}{2} \implies w_1 = 2 \left[\cos\left(\frac{\pi}{2}\right) + i \sin\left(\frac{\pi}{2}\right) \right] = 2(0 + i) = 2i$$

- For $k = 2$:

$$\theta_2 = -\frac{\pi}{6} + \frac{4\pi}{3} = \frac{7\pi}{6} \implies w_2 = 2 \left[\cos\left(\frac{7\pi}{6}\right) + i \sin\left(\frac{7\pi}{6}\right) \right] = 2 \left(-\frac{\sqrt{3}}{2} - \frac{1}{2}i \right) = -\sqrt{3} - i$$

4. Quick Algebraic Check:

$$(2i)^3 = 8i^3 = 8(-i) = -8i \quad \checkmark$$

$$(\sqrt{3} - i)^3 = (\sqrt{3})^3 - 3(\sqrt{3})^2(i) + 3(\sqrt{3})(i^2) - i^3 = 3\sqrt{3} - 9i - 3\sqrt{3} + i = -8i \quad \checkmark$$

- **Exemplar Problem 2 (Second-Order Initial Value Problem):** Solve the initial value problem $y'' + 4y' + 13y = 0$, $y(0) = 2$, $y'(0) = -1$.

• Analytical Step-by-Step Derivation:

1. Form the auxiliary (characteristic) equation:

$$\lambda^2 + 4\lambda + 13 = 0$$

2. Compute roots via quadratic formula:

$$\lambda = \frac{-4 \pm \sqrt{4^2 - 4(1)(13)}}{2} = \frac{-4 \pm \sqrt{16 - 52}}{2} = \frac{-4 \pm \sqrt{-36}}{2} = \frac{-4 \pm 6i}{2} = -2 \pm 3i$$

3. Identify real and imaginary parameters:

$$\alpha = -2, \quad \beta = 3$$

4. Form the general oscillatory solution:

$$y(x) = e^{-2x} (C_1 \cos(3x) + C_2 \sin(3x))$$

5. Apply initial condition $y(0) = 2$:

$$y(0) = e^0 (C_1 \cos(0) + C_2 \sin(0)) = C_1 = 2$$

6. Compute derivative $y'(x)$ using product rule:

$$y'(x) = -2e^{-2x} (C_1 \cos(3x) + C_2 \sin(3x)) + e^{-2x} (-3C_1 \sin(3x) + 3C_2 \cos(3x))$$

7. Apply initial condition $y'(0) = -1$:

$$y'(0) = -2(C_1) + 3(C_2) = -2(2) + 3C_2 = -4 + 3C_2 = -1$$

$$3C_2 = 3 \implies C_2 = 1$$

8. State the unique explicit particular solution:

$$y(x) = e^{-2x} (2 \cos(3x) + \sin(3x))$$